

Multiple Choice

1. **Answer:** A

Options: A) $\text{PbH}_4 < \text{SnH}_4 < \text{GeH}_4 < \text{SiH}_4 < \text{CH}_4$; B) $\text{PbH}_4 < \text{SnH}_4 < \text{CH}_4 < \text{GeH}_4 < \text{SiH}_4$; C) $\text{CH}_4 < \text{SiH}_4 < \text{GeH}_4 < \text{SnH}_4 < \text{PbH}_4$; D) $\text{CH}_4 < \text{PbH}_4 < \text{GeH}_4 < \text{SnH}_4 < \text{SiH}_4$

Note: Correct option: A - $\text{PbH}_4 < \text{SnH}_4 < \text{GeH}_4 < \text{SiH}_4 < \text{CH}_4$

2. **Answer:** B

Options: A) 23.56 GHz; B) 42.58 MHz; C) 74.34 kHz; D) 13.93 MHz

Note: Correct option: B - 42.58 MHz

3. **Answer:** A

Options: A) MgO; B) MgS; C) NaF; D) NaCl

Note: Correct option: A - MgO

4. **Answer:** A

Options: A) It has little effect.; B) It significantly increases the pH.; C) It significantly decreases the pH.; D) It changes the pH asymptotically to the pKa of the acid.

Note: Correct option: A - It has little effect.

5. **Answer:** B

Options: A) NH_3 ; B) NH_2^- ; C) NH_4^+ ; D) N_2H_4

Note: Correct option: B - NH_2^-

True / False

6. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

7. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

8. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

9. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'The atomic radii of the lanthanide elements increase across the period from La to Lu.'

10. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'The MnO₄⁻ : I⁻ ratio is 6:5.'

Long Form Response

11. **Answer:** CH₃Cl. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** 1:8:28:56:70:56:28:8:1. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key